Rhilo Sotto

COMPE470

21 October 2024

Assignment 2

Modules

Datapath

```
module Datapath #(parameter WL = 4)
(input CLK, input signed [WL - 1: 0] a, b, c, X,
output signed [4*WL - 1: 0] out);

reg signed [2*WL - 1: 0] R0, R1, R2;
reg signed [4*WL - 1: 0] R3;

  always @(posedge CLK) begin
    R0 <= (a * X) + b;
    R1 <= X;
    R2 <= c;

    R3 <= (R0 * R1) + R2;
  end

  assign out = R3;

endmodule
```

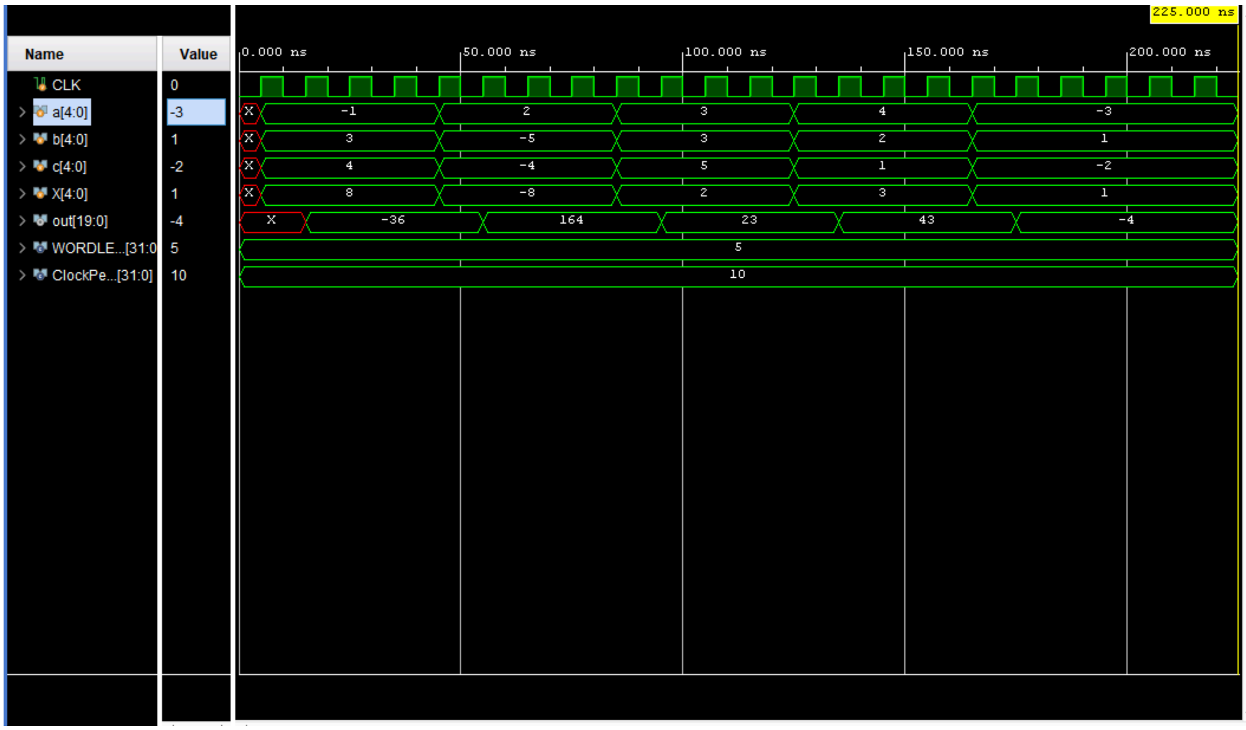
Testbenches

Datapath

```
`timescale 1ns / 1ns


module tb_datapath;

localparam WORDLENGTH = 5;

reg CLK;
reg signed [WORDLENGTH-1:0] a, b, c, X;
wire signed [4*WORDLENGTH-1:0] out;

// clock signal
parameter ClockPeriod = 10;
initial CLK = 0;
always #(ClockPeriod / 2) CLK = ~CLK;

// instantiate DUT with parameter
Datapath #(.WL(WORDLENGTH)) FD0 (.CLK(CLK), .a(a), .b(b), .c(c), .X(X), .out(out));

initial begin
   @(posedge CLK)     a = -1; b =  3; c =  4; X =  8;
   #(ClockPeriod * 4) a =  2; b = -5; c = -4; X = -8;
   #(ClockPeriod * 4) a =  3; b =  3; c =  5; X =  2;
   #(ClockPeriod * 4) a =  4; b =  2; c =  1; X =  3;
   #(ClockPeriod * 4) a = -3; b =  1; c = -2; X =  1;
   #(ClockPeriod * 6) $finish;

end


endmodule
```

Waveforms

Datapath



Corresponding to the datapath module and testbench, the output is as expected.